

MANIPAL INSTITUTE OF TECHNOLOGY

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**Mini Project Report
of
Database Management System Lab (MCA 4141)**

TOY MANUFACTURING AND IMPORT DEPENDENCY ANALYSIS SYSTEM

SUBMITTED BY

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CERTIFICATE

This is to certify that the project titled “Toy Manufacturing and Import Dependency Analysis System” is a record of the bona fide work done by Student(s) (Reg. No. _____) submitted in partial fulfilment of the requirements for the award of the Degree of Master of Computer Applications of Manipal Institute of Technology, Manipal, Karnataka, during the academic year 2026–2027.

Name	and	Signature	of	Examiners:
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ABSTRACT

The Toy Manufacturing and Import Dependency Analysis System is a database-oriented mini project designed to manage and analyze the lifecycle of toy manufacturing activities. The system maintains structured information about toys, raw materials, suppliers, sourcing options, manufacturing batches, inventory, sales channels, and sales transactions. The proposed database supports the operational flow from raw material sourcing to toy manufacturing, inventory maintenance, sales, and profit analysis.

The distinctive feature of the proposed system is its Import Dependency and Local Substitution Impact Analysis. The system identifies raw materials sourced through imported suppliers, determines whether locally available alternatives exist, and compares their costs. This allows the manufacturer to evaluate the possible impact of local substitution on manufacturing cost and potential profitability. The system does not assume that local sourcing is always cheaper; instead, it provides a data-driven comparison.

The project is intended for implementation using Oracle SQL and PL/SQL and demonstrates relational database design, normalization, constraints, joins, aggregate functions, nested queries, views, indexes, sequences, stored procedures, functions, and triggers. The scope is intentionally limited so that it remains a feasible DBMS mini project rather than a complete enterprise resource planning system.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Toy manufacturing involves interconnected activities such as raw material procurement, supplier selection, production, stock management, and sales. When these activities are maintained through separate records, it becomes difficult to obtain a consolidated view of production cost, sourcing information, inventory, and sales performance.

1.2 Interpretation of the Original Abstract

The original abstract requires a system that can provide information about toys, imported sources, raw materials, manufacturing stages, e-commerce sales, profit analysis, and reduction of imported goods. If interpreted as a complete business application, this would require supply-chain management, procurement, logistics, accounting, and external platform integration. Such a system would be too large for a DBMS mini project. Therefore, the proposed project refines the abstract into a focused relational database and decision-support system.

1.3 Need for the Proposed System

- Centralized storage of toy, material, supplier, production, inventory, and sales data.
- Traceability from raw material sourcing to finished toy sales.
- Identification of imported and local sourcing options.
- Manufacturing cost and profit analysis through SQL queries.
- A practical method to analyze local alternatives for imported materials.

1.4 Scope of the Project

The project is organized into four integrated modules: (1) Toy & Material Management, (2) Supplier & Sourcing Management, (3) Manufacturing & Inventory Management, and (4) Sales & Business Analysis. Together, these modules cover raw material sourcing, toy requirements, production batches, inventory, sales, profit analysis, and import dependency analysis. The project excludes real-time e-commerce APIs, online payments, customer management, logistics tracking, customs processing, demand forecasting, machine learning, and complete ERP functionality.

CHAPTER 2: PROBLEM STATEMENT & OBJECTIVES

2.1 Problem Statement

Toy manufacturers require a structured mechanism for maintaining information about toys, raw materials, suppliers, manufacturing activities, inventory, and sales. Dependence on imported raw materials may influence production cost and sourcing decisions. When sourcing information is not integrated with toy requirements and manufacturing records, it becomes difficult to determine which toys depend on imported materials, whether local alternatives exist, and how a sourcing change may affect cost and profit.

2.2 Objectives

1. Maintain information about toys and their categories.
2. Maintain a master database of raw materials.
3. Module 2: Maintain supplier information and multiple sourcing options for the same material.
4. Module 2: Classify sourcing options as LOCAL or IMPORTED and store country of origin, cost, and availability.
5. Classify sourcing options as LOCAL or IMPORTED.
6. Record country of origin, unit cost, and availability of supplied materials.
7. Track manufacturing batches and actual material usage.
8. Maintain current finished toy inventory.
9. Record sales through multiple sales channels.
10. Calculate manufacturing cost, revenue, and profit.
11. Analyze import dependency and local substitution opportunities.
12. Compare imported and local sourcing costs.

2.3 Existing System

A simplified existing approach may rely on spreadsheets, manual registers, and disconnected files. Supplier data, production records, stock, and sales information may be maintained separately. This can lead to data redundancy, inconsistent updates, difficulty in tracing relationships, and manual calculation of cost and profitability.

2.4 Proposed System

The proposed system integrates the manufacturing lifecycle through four consolidated database modules rather than treating every business activity as a separate module. Relational tables and foreign keys connect suppliers, raw materials, toys, manufacturing batches, inventory, sales channels, and sales transactions. SQL queries and views provide analytical reports, while selected PL/SQL procedures, functions, and triggers implement useful business logic.

2.5 Novelty / Unique Feature

The Import Dependency and Local Substitution Impact Analysis is the project's principal novelty. A raw material can have multiple supplier-material records. Some may be imported and others local. The system compares available sourcing options for the same material and estimates the resulting cost difference when multiplied by the quantity required for a toy. This makes the project a database-based decision-support system rather than only an inventory or manufacturing management system.

CHAPTER 3: FUNCTIONAL REQUIREMENTS – MODULES AND SOFTWARE USED

3.1 Modules of the System

MODULE	OBJECTIVE	TABLES	OUTPUT
1. Toy & Material Management	Maintain toy information, raw material details, and the material quantities required to manufacture each toy.	TOY, RAW_MATERIAL, TOY_MATERIAL	Toy master list, material master list, and toy-wise material requirement report
2. Supplier & Sourcing Management	Maintain suppliers, material sourcing options, source type, country of origin, availability, and cost. This module supports import dependency and local substitution analysis.	SUPPLIER, SUPPLIER_MATERIAL	Supplier-wise supply report, imported versus local sourcing report, and local substitution comparison
3. Manufacturing & Inventory Management	Track toy production batches, actual material usage, production costs, and current finished toy stock.	MANUFACTURING_BATCH, BATCH_MATERIAL_USAGE, INVENTORY	Manufacturing history, batch cost report, current inventory, and low-stock report
4. Sales & Business Analysis	Record sales through different channels and generate revenue, profit, and sourcing-impact analysis using related production and sourcing data.	SALES_CHANNEL, SALE (with analytical use of related tables)	Channel-wise sales, toy-wise revenue, profit analysis, most profitable toys, and import dependency impact reports

3.2 Functional Requirements

- Module 1: Add, update, and retrieve toy and raw material records.
- Module 1: Define and maintain toy-wise raw material requirements.
- Map multiple materials to a toy through a bill-of-materials relationship.
- Module 3: Record manufacturing batches, actual material usage, production costs, and current inventory.
- Module 4: Record sales through multiple channels and generate revenue and profit reports.
- Module 4: Generate analytical reports comparing imported sourcing with available local alternatives.

3.3 Non-Functional Requirements

- Data integrity through primary keys, foreign keys, and CHECK constraints.
- Normalized design to reduce redundancy.
- Implementation using Oracle SQL and PL/SQL.
- Simple and maintainable structure suitable for academic demonstration.
- No dependence on external APIs or advanced infrastructure.

3.4 Hardware and Software Requirements

CATEGORY	REQUIREMENT
Operating System	Windows or another OS supported by Oracle development tools
Database	Oracle Database / Oracle XE where appropriate
Development Tool	Oracle SQL Developer or equivalent
Languages	SQL and PL/SQL
Documentation	Microsoft Word or compatible software
Hardware	Standard laptop/desktop capable of running Oracle development tools

3.5 Overall System Workflow

Supplier / Source → Raw Material → Toy Material Requirement → Manufacturing Batch → Inventory → Sales Channel and Sale → Profit Analysis. The novelty feature is integrated primarily into Module 2 (Supplier & Sourcing Management) and Module 4 (Sales & Business Analysis): sourcing data identifies imported dependence and available local alternatives, while analytical reports evaluate their possible impact on manufacturing cost and profitability.

